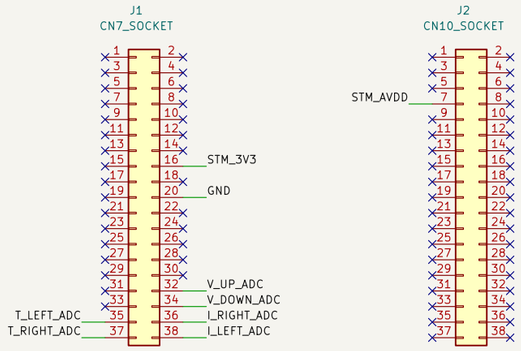
TMR MONITOR SHIELD | RevA.0 FUNCTIONAL LAYOUT

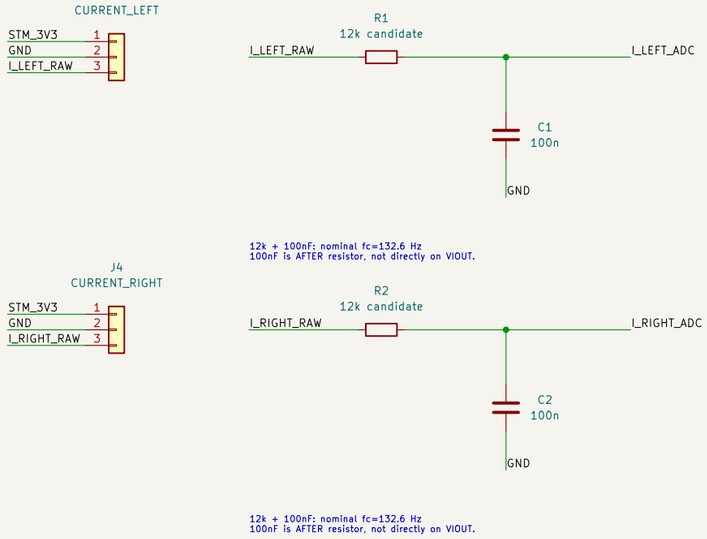
DRAFT — NOT FOR FABRICATION OR CONNECTION
Signal circuits only: current-sensor power path remains off-board.

01 NUCLEO-F446RE INTERFACE



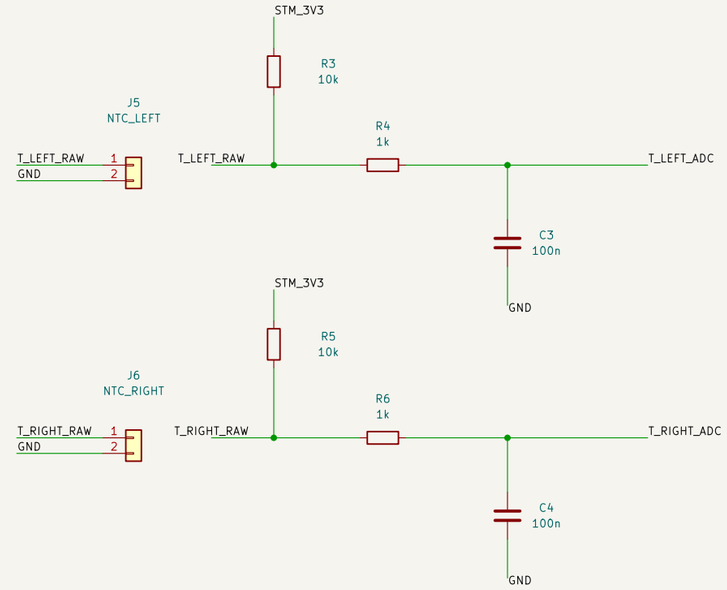
CN7: 16=3V3, 20=GND; 32=PA4, 34=PB0
35=PC2, 37=PC3; 36=PC1/PB9*, 38=PC0/PB8*
* Verify solder bridges; preserve IMU PB8/PB9.
CN10: 7=AVDD (TPS / TMUX supply). Other pins unused here.
NC marks do not alter the NUCLEO wiring.

02 CURRENT INPUTS



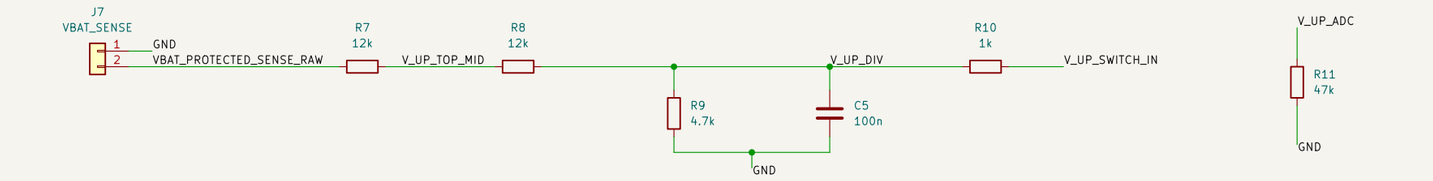
12k + 100nF: nominal fc=132.6 Hz
100nF is AFTER resistor, not directly on VIOUT.

03 NTC INPUTS

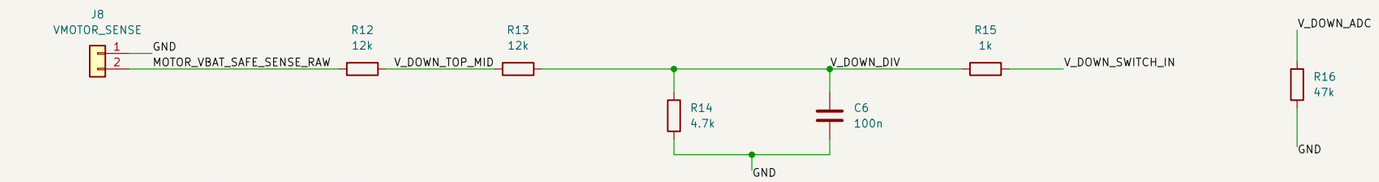


04 VOLTAGE INPUTS — TWO CHANNELS

BATTERY SIDE | J7 / R7–R11 / C5

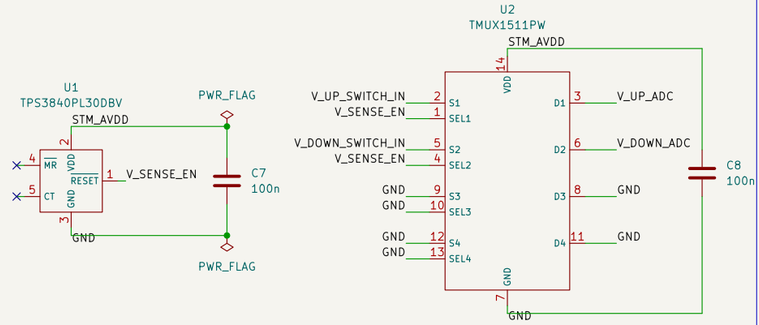


MOTOR SUPPLY SIDE | J8 / R12–R16 / C6



Two voltage channels connected to PA4 / PB0. Connection contract 27; ERC checked 2026-11-07.

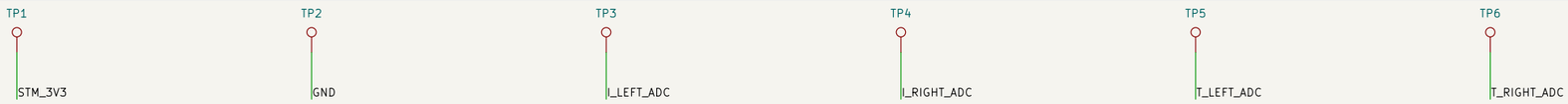
05 VOLTAGE SUPERVISOR / SWITCH



OPEN ELECTRICAL ITEMS
- Port ESD / misconnection / transient protection remains to be designed.
- TPS / TMUX connected; power-down backfeed behavior remains unverified.
- Supply margin, sensor cable capacitance, ADC leakage and sampling / filtering remain open.
- No automatic motor stop threshold is defined by this sheet.

OPEN MECHANICAL ITEMS
- Connector models and resistor/capacitor body envelopes are provisional.
- Template header positions only; actual socket height / clearance not verified.
- Keep RESET / USER / jumpers / ST-LINK access and cable strain relief.
- A clean ERC is not proof of these electrical or mechanical conditions.

06 TEST POINTS



NOT FOR FABRICATION — protection / mechanical review open

Sheet: /	
File: TMR_Monitor_Shield_RevA.kicad_sch	
Title: TMR Monitor Shield	
Size: A3	Date: 2026-11-07
KiCad E.D.A. 10.0.5	**Rev: RevA.1-DRAFT**
	Id: 1/1